ASSIGNMENT PROBLEM – ONE PAGE NOTES

Definition:
Assignment Problem is a special case of Transportation Problem where each person is assigned to exactly one job such that total cost/time is minimum (or profit is maximum).

Key Characteristics:
- One person → One job
- One job → One person
- Supply = 1 for each row
- Demand = 1 for each column
- No splitting of assignments

Mathematical Formulation:
Minimize $Z = \Sigma\Sigma\, c_{ij}\, x_{ij}$
Subject to:
$\Sigma\, x_{ij} = 1$ (each person gets one job)
$\Sigma\, x_{ij} = 1$ (each job done by one person)
$x_{ij} = 0$ or 1

Solution Method:
Solved using Hungarian Method.

Hungarian Method Steps:
1. Row reduction
2. Column reduction
3. Cover zeros with minimum lines
4. Check optimality (lines = order of matrix)
5. Modify matrix if needed
6. Make final assignments

Maximization Assignment Problem:
Convert to minimization by:
New value = Maximum value – Given value

Unbalanced Assignment Problem:
When number of persons ≠ number of jobs.
Solution: Add dummy row/column with zero cost to balance.

Assignment vs Transportation Problem:
- Assignment is a special case of TP
- TP allows splitting, AP does not
- TP uses NWCM/LCM/VAM
- AP uses Hungarian Method

Important Exam Point:
Assignment Problem is a Transportation Problem with unit supply and unit demand.